

The landscape, not the language, decides which search method fits

Smooth and single-valley

use gradient descent or L-BFGS-B



Cheap gradient

few evaluations, fast convergence

Rugged or many valleys

use a population or annealing search



Many evaluations

budget and random seed must be reported

Noisy or non-numeric

use a derivative-free metaheuristic



Always benchmark

random search is the honest baseline